[T]-Theory: Mathematics

P4 · Formal Foundation · For: The Type Theorist

Functorial Green's Function — the proof architecture in Lean 4.

Build status (2026-08-15):

- `lake build`: 3912/3912 jobs, exit 0
- 22 defaultTargets, all **clean** (warnings only)
- 0 Float in proof files (ISS-009 closed)
- 2 bare `sorry`s remain (Hopfield, ISS-011)

Gate theorem architecture:

- `LocalGR.lean`: linearised GR gate
- `LocalGeometry.lean`: compact geometry gate
- `calabi_yau_moduli_static`: axiom → **theorem**
- `dm_gauge_coupling_zero`: axiom → **theorem**

I · The Functorial Structure

The USF propagator G is a **natural transformation**:

$$G : \mathrm{Scale}_{20} \to \mathrm{FieldEq}$$

where $\mathrm{Scale}_{20} \simeq$ Fin 21 and $\mathrm{FieldEq}(\sigma)$ is the type of field configurations at scale σ.

Scale invariance as a coherence condition:

$$G \circ Z_\sigma = Z_{\sigma+1} \circ G \quad \forall\, \sigma$$

The Zoom Operator Z_σ is the functor between adjacent fibers. Proved in Lean 4 via `scale_invariance_full`.

II · The (□)-Type Decomposition

$$\mathrm{SomaField} \;\equiv\; \sum_{(\sigma\,:\,\mathrm{Fin}\ 21)} \mathrm{Substrate}(\sigma)$$

Each $\mathrm{Substrate}(\sigma)$: Type is the Lean 4 type of the physical carrier at scale σ (e.g. `Fin 3 →` for M_3, `CompactX7` for X_7).

Lean types used: `Matrix (Fin 8) (Fin 8)` (coupling W_8), `RigidAttractor V` (moduli stability), `G2CompactManifold` (compact geometry).

III · Honest Axiom Accounting

All 11-dimensional claims bottom out in precisely-typed local axioms — no vacuous `True` wildcards remain:

- `g2_holonomy_implies_rigid_attractor` — Berger classification (pending Mathlib)
- `rigidAttractor_freezes_omega_lambda` — GR perturbation theory (pending Mathlib)
- `g2_implies_hw_compactification` — Horava-Witten geometry (pending Mathlib)

Each axiom names its exact Mathlib dependency. The proof tree is auditable.

IV · Open Proof Obligations

- **ISS-011**: Upgrade `Pattern = Fin D →` to `SpinState` for Hopfield convergence
- **ISS-012**: Add `lean-appendix` to `lake build`
- **ISS-014**: Path-dependence in moduli space (needs ISS-016)
- **Mathlib**: Riemannian holonomy, GR perturbation theory, KK spectral theory

G-ID: *Functorial Green's Function* — category-theoretic propagator between field spaces. ORCID: 0009-0007-2194-0850 · · Zurich 2026